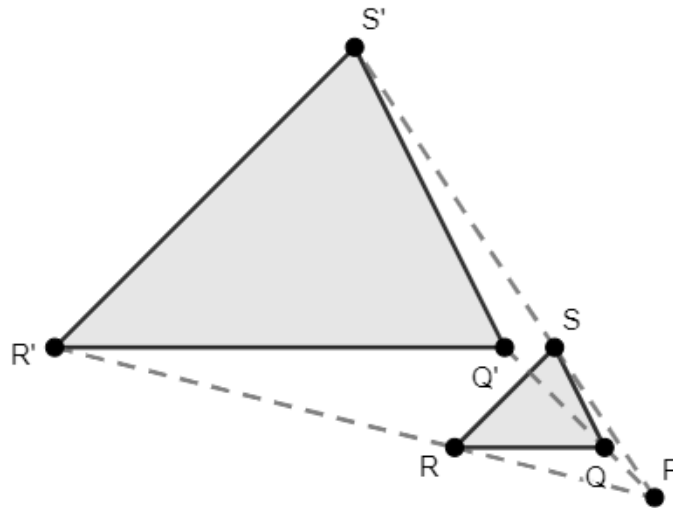


$\triangle Q'R'S'$ was created by dilating $\triangle QRS$. The length of $\overline{QR}$ is 3 and the length of $\overline{Q'R'}$ is 9.



- $\triangle QRS$ is the

☐ image
☐ preimage

, $\triangle Q'R'S'$ is the

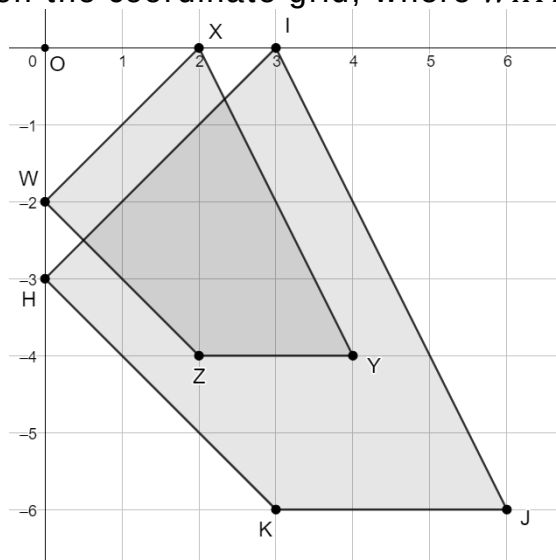
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☐ preimage

, and the center of dilation is point

☐ P
☐ Q
☐ Q'

.
- Determine the dilation ratio.
- If the length of $\overline{Q'S'}$ is $3\sqrt{5}$, determine the length of $\overline{QS}$.
- What is the ratio of $\frac{RS}{R'S'}$?
- What is the ratio of $\frac{P'R'}{PR}$?

$WXYZ$ and $HIJK$ are shown on the coordinate grid, where $WXYZ$ is a dilation of $HIJK$.

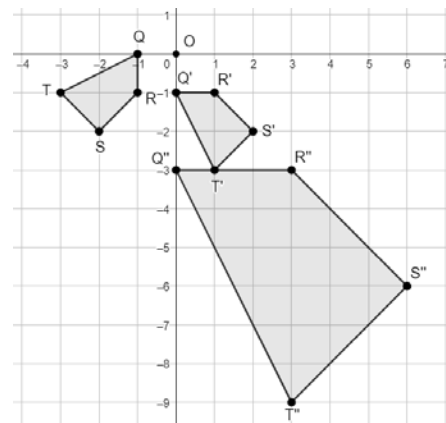


6. Complete the table.

Segment	Slope	Distance
$\overline{YZ}$		
$\overline{JK}$		
$\overline{WX}$		
$\overline{HI}$		

7. Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.
8. Write a description of the dilation by giving the center of dilation and the scale factor.
9. What is the ratio $\frac{OX}{OI}$?
10. What is the ratio $\frac{OX}{XI}$?

A sequence of transformations of $QRST$ is shown on the coordinate grid. $QRST$ is rotated then dilated, where $Q'R'S'T'$ is the rotation and $Q''R''S''T''$ is the dilation of $Q'R'S'T'$.



Complete the descriptions:

- $QRST$ was _____ to create $Q'R'S'T'$.
- $Q'R'S'T'$ was dilated using a scale factor of _____ centered at _____ to create $Q''R''S''T''$.
- Complete the table:

Transformation	Algebraic Description
Rotation	$(x,y) \rightarrow$
Dilation	$(x,y) \rightarrow$

- Determine the coordinates if $QRST$ is first dilated then rotated:

Preimage $QRST$	
Q	$(-1,0)$
R	$(-1,-1)$
S	$(-2,-2)$
T	$(-3,-1)$

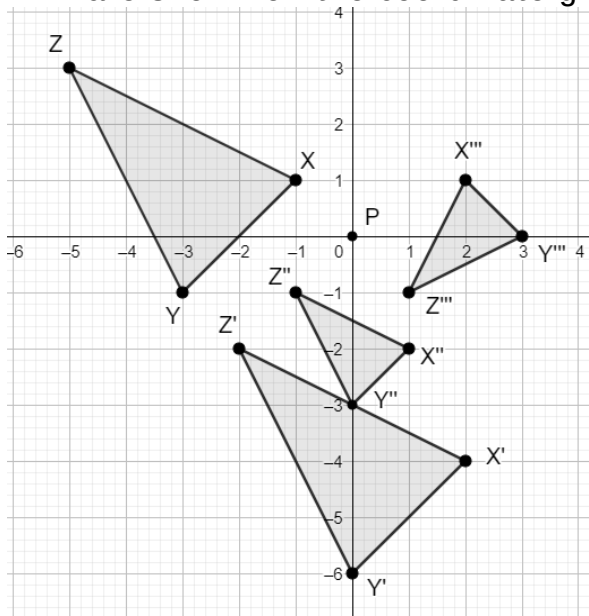
Dilation $Q'R'S'T'$	
Q'	
R'	
S'	
T'	

Rotation $Q''R''S''T''$	
Q''	
R''	
S''	
T''	

- Based on the results above, the sequence of transformations commutative.

☐ is
☐ is not

$XYZ, X'Y'Z', X''Y''Z'',$ and $X'''Y'''Z'''$ are shown on the coordinate grid.



6. Determine what transformations occurred and in what order:
- XYZ is first

☐ translated
☐ reflected
☐ rotated
☐ dilated

 to $X'Y'Z'$, then

☐ translated
☐ reflected
☐ rotated
☐ dilated

 to $X''Y''Z''$,
- and finally

☐ translated
☐ reflected
☐ rotated
☐ dilated

 to $X'''Y'''Z'''$.
7. Describe the transformation for $XYZ \rightarrow X'Y'Z'$.
8. Describe the transformation for $X'Y'Z' \rightarrow X''Y''Z''$.
9. Describe the transformation for $X''Y''Z'' \rightarrow X'''Y'''Z'''$.

10. Write the algebraic description for the sequence of transformations from 7-9:

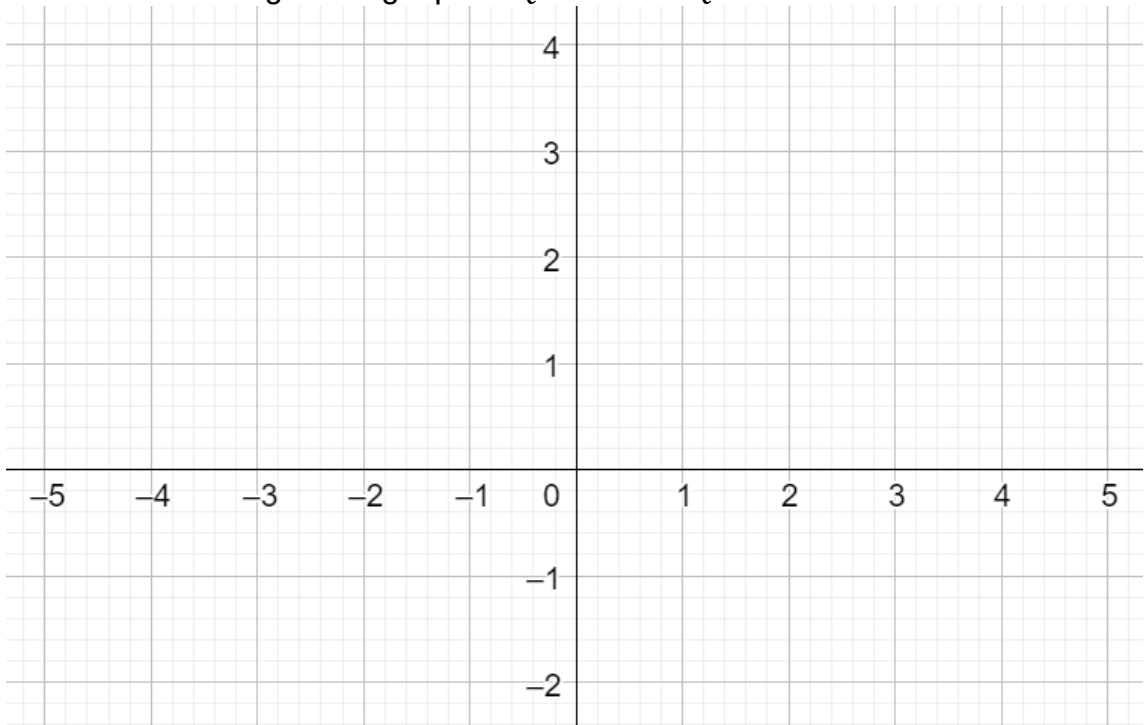
Transformation	Algebraic Description
	$(x,y) \rightarrow$
	$(x,y) \rightarrow$
	$(x,y) \rightarrow$

The coordinates of $\triangle QRS$ and $\triangle Q'R'S'$ are shown in the table, where $\triangle QRS$ is the preimage:

$\triangle QRS$		
Vertex	x	y
Q	-1	2
R	-3	1
S	-4	4

$\triangle Q'R'S'$		
Vertex	x	y
Q'	2	1
R'	1	3
S'	4	4

1. Use the coordinate grid to graph $\triangle QRS$ and $\triangle Q'R'S'$.



2. What type of transformation was applied to the preimage $\triangle QRS$?
3. Was distance preserved in this transformation?
4. $\triangle Q'R'S'$ is further transformed by dilating with a scale factor of $\frac{1}{2}$ centered at the origin to create $\triangle Q''R''S''$. Graph $\triangle Q''R''S''$ on the coordinate grid above.
5. Was distance preserved in this second transformation from $\triangle Q'R'S'$ to $\triangle Q''R''S''$?

Complete the following statements about the transformation of preimage XYZ :

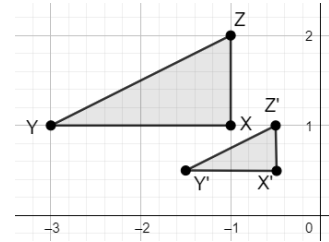
6. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

This transformation

- ☐ does
- ☐ does not

preserve distance.



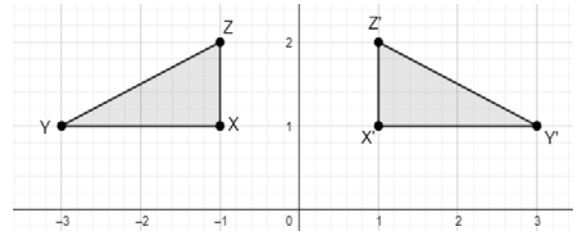
7. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

This transformation

- ☐ does
- ☐ does not

preserve distance.



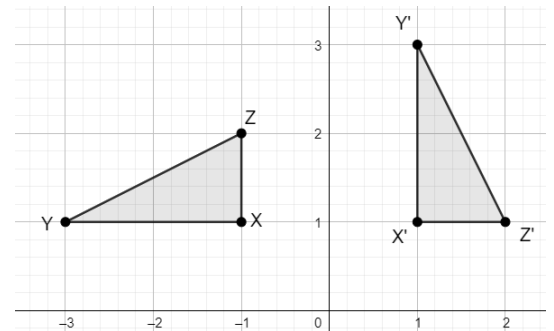
8. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

This transformation

- ☐ does
- ☐ does not

preserve distance.



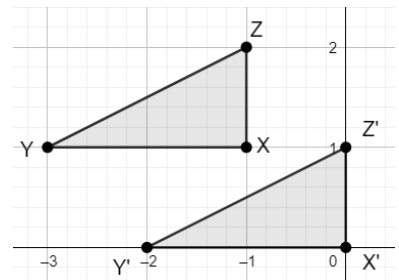
9. Image $X'Y'Z'$ was created by

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

This transformation

- ☐ does
- ☐ does not

preserve distance.



10. Image MLH was created either by

or by

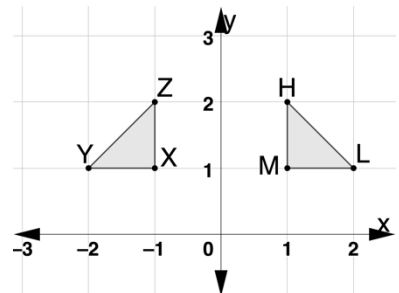
- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

- ☐ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

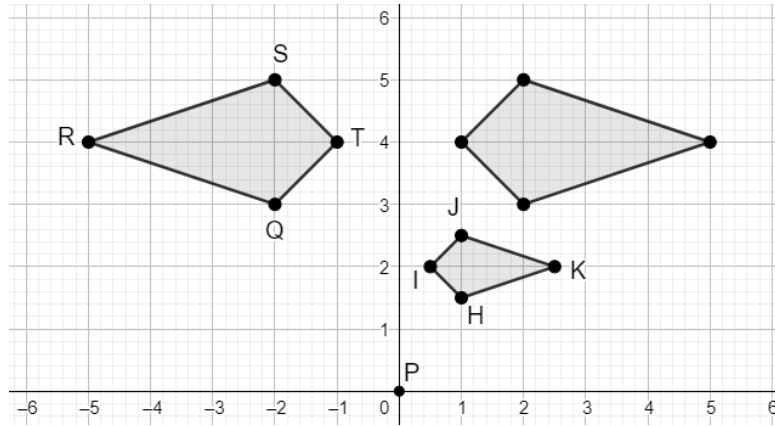
Both of these transformations

- ☐ do
- ☐ do not

preserve distance.

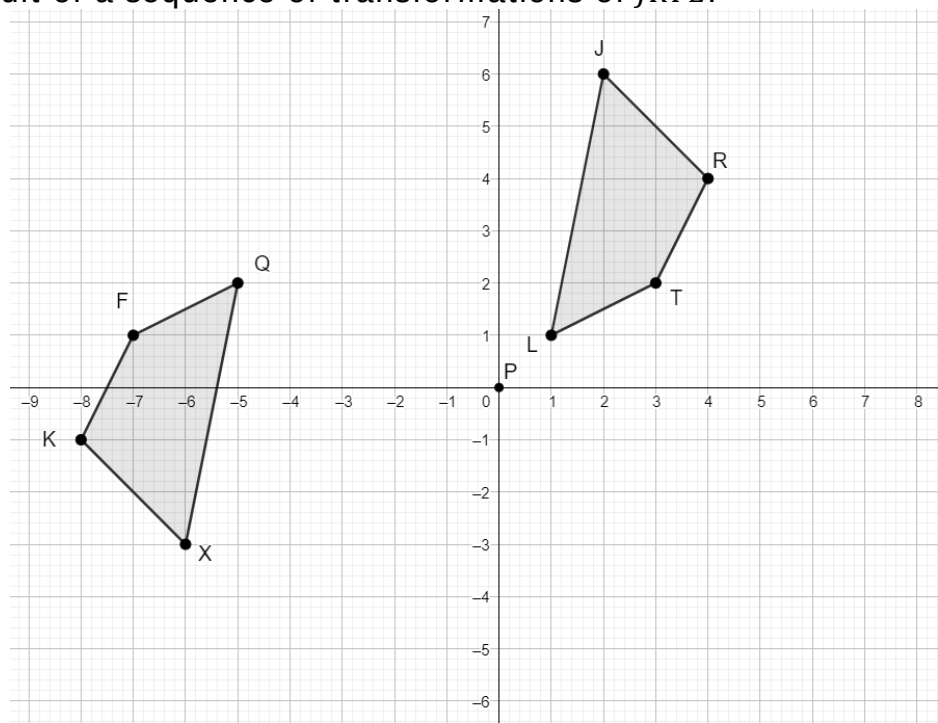


$HIJK$ is the result of a sequence of transformations of $QTSR$. The unlabelled quadrilateral is the first transformation.



1. Describe the first transformation of $QTSR$.
2. Write the transformation algebraically.
3. Describe the second transformation that results in $HIJK$.
4. Write the second transformation algebraically.
5. Explain if the sequence of transformations preserves distance.

XKFQ is the result of a sequence of transformations of *JRTL*.



Julia and Nathan were determining the sequence of transformations, but each came up with a different sequence. Complete the table for the missing portions of their transformations.

			First Transformation	Second Transformation
6.	Julia's Sequence	Written Description		Rotate 180° about the origin.
7.		Algebraic Description	$(x,y) \rightarrow (x + 4,y - 3)$	
8.	Nathan's Sequence	Written Description	Rotate 180° about the origin.	
9.		Algebraic Description		$(x,y) \rightarrow (x + 4,y - 3)$

10. Considering both proposed sequences, you conclude that correct.

- ☐

 Julia's sequence is
- ☐
- Nathan's sequence is

☐

Geometry

Unit 6

Lesson 5 Homework

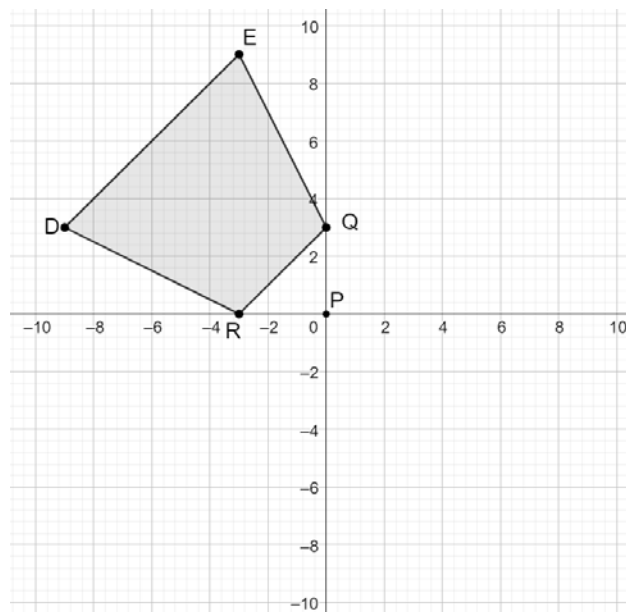
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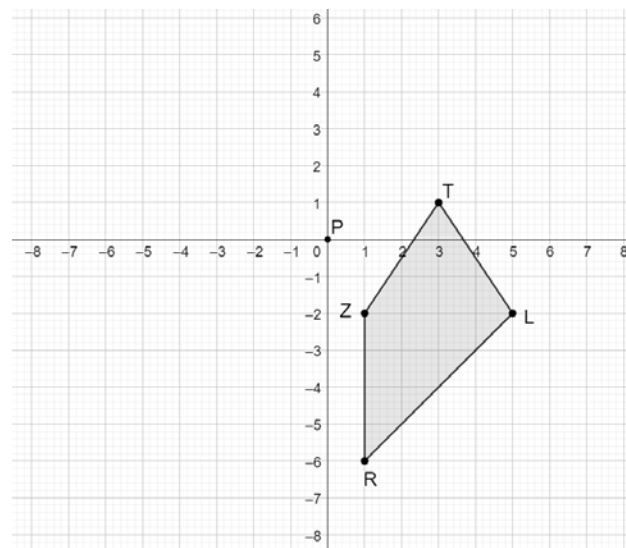
$QEDR$ and point P are shown on the coordinate plane.

1. Draw the quadrilateral that is a 180° rotation of $QEDR$ about the origin. Label the quadrilateral $Q'E'D'R'$.
2. Draw the quadrilateral that is a dilation of $Q'E'D'R'$ with a center of dilation at point P and a scale factor of $\frac{2}{3}$. Label the quadrilateral $Q''E''D''R''$.



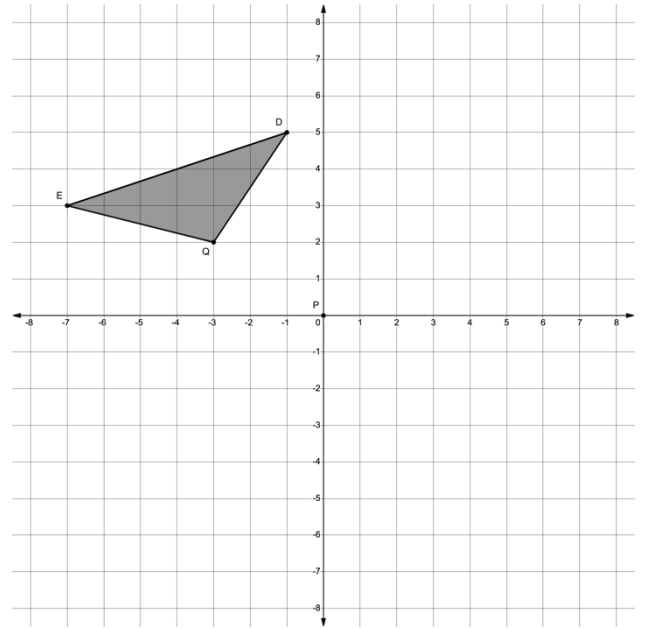
$ZTLR$ and point P are shown on the coordinate plane.

3. Draw the quadrilateral that is a dilation of $ZTLR$ with a center of dilation at point Z and a scale factor of $\frac{3}{2}$. Label the quadrilateral $Z'T'L'R'$.
4. Draw the quadrilateral that is a reflection of $Z'T'L'R'$ across the y -axis. Label the quadrilateral $Z''T''L''R''$.



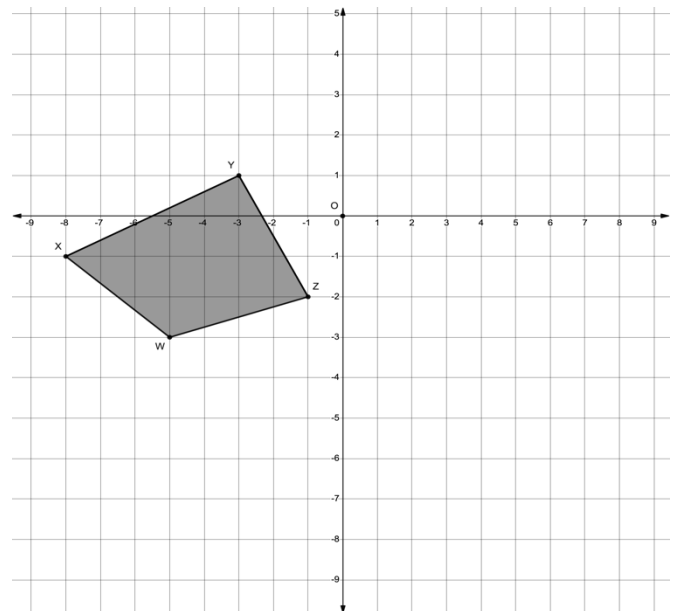
$\triangle QED$ and point P are shown on the coordinate plane.

5. Draw the triangle that is a 90° clockwise rotation of $\triangle QED$ about point P . Label the quadrilateral $\triangle Q'E'D'$.
6. Draw the triangle that is a reflection of $\triangle Q'E'D'$ across the x -axis. Label the quadrilateral $\triangle Q''E''D''$.



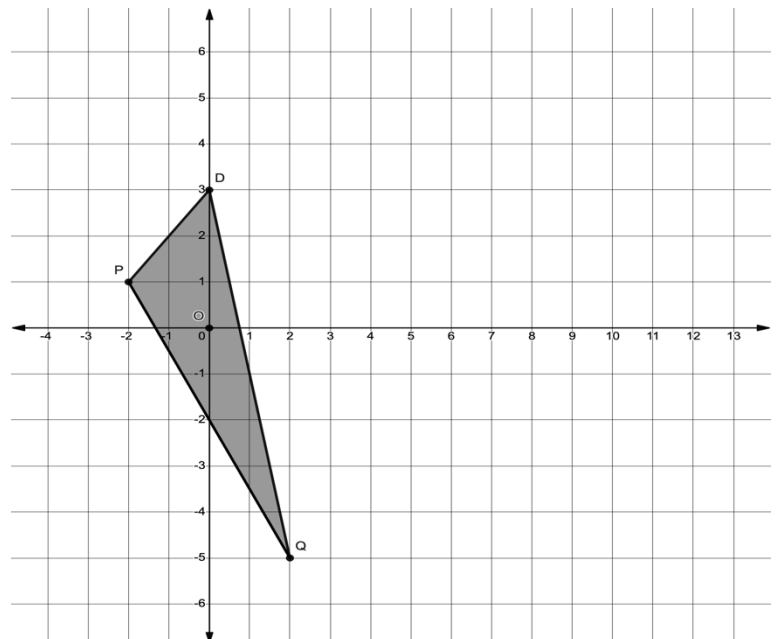
$WXYZ$ and point O are shown on the coordinate plane.

7. Draw the quadrilateral that is a 90° counterclockwise rotation of $WXYZ$ about the origin. Label the quadrilateral $W'X'Y'Z'$.
8. Draw the quadrilateral that is a translation of $W'X'Y'Z'$ 2 units right and 4 units up. Label the quadrilateral $W''X''Y''Z''$.



$\triangle PDQ$ and point O are shown on the coordinate plane.

9. Draw $\triangle P'D'Q'$ as a result of $(x, y) \rightarrow (x + 6, y + 1)$.
10. Draw $\triangle P''D''Q''$ as a result of $(x, y) \rightarrow \left(\frac{3}{2}x, \frac{3}{2}y\right)$.



1. When a sequence of transformations results in both angles and sides of the image that are congruent to corresponding angles and sides of the preimage, the

sequence

- ☐ does
☐ does not

include a dilation. If it

- ☐ does
☐ does not

include a dilation,

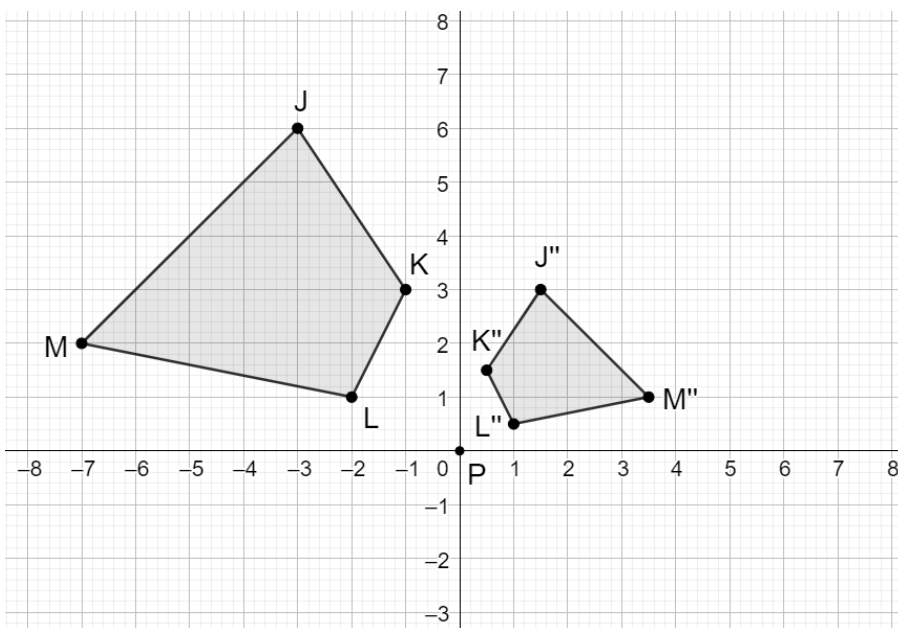
the corresponding

- ☐ angles
☐ sides

are congruent while the corresponding

- ☐ angles
☐ sides

are proportional.

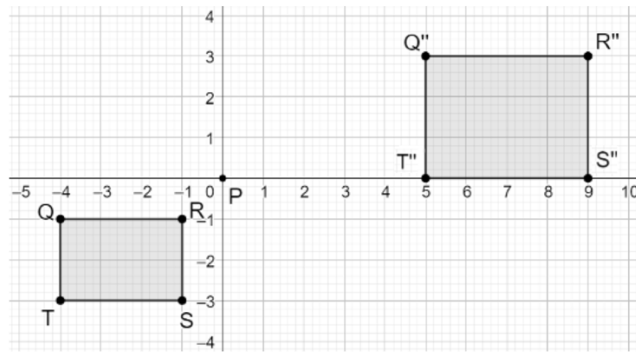


2. Write a sequence of transformations for preimage $JKLM$ that would show whether the figures are similar.

3. Is $J''K''L''M''$ similar to the primage?

- ☐ Yes
☐ No

4. Explain the reason for your answer for #3.

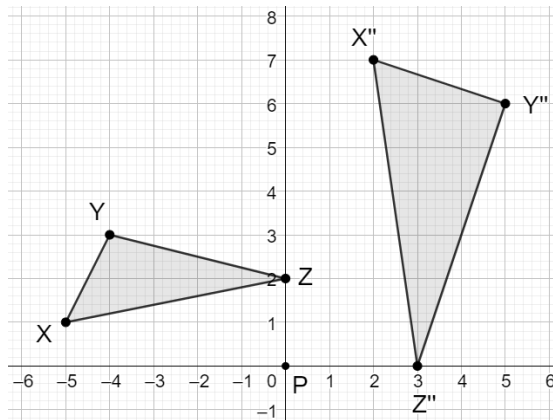


5. Write a sequence of transformations for preimage $QRST$ that would show whether the figures are similar.

6. Is $Q''R''S''T''$ similar to the preimage?

- ☐ Yes
☐ No

7. Explain the reason for your answer for #6.



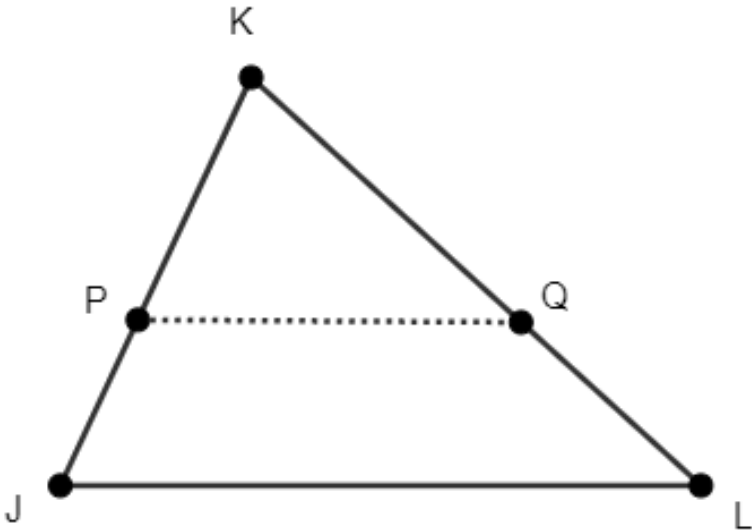
8. Write a sequence of transformations for preimage $\triangle XYZ$ that would show whether the figures are similar.

9. Is $\triangle X''Y''Z''$ similar to the preimage?

- ☐ Yes
☐ No

10. Explain the reason for your answer for #9.

$\triangle KJL$ is shown, where $m\angle KJL = 65^\circ$, $m\angle KLJ = 42^\circ$, and $m\angle JKL = 73^\circ$. Point P is on $\overline{KJ}$ and Point Q is on $\overline{KL}$. $\overline{PQ}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle JKL$		Measurements of $\triangle PKQ$	
$JK =$	$m\angle KJL = 65^\circ$	$PK =$	$m\angle KPQ =$
$KL =$	$m\angle JKL = 73^\circ$	$KQ =$	$m\angle PKQ = 73^\circ$
$JL =$	$m\angle KLJ = 42^\circ$	$PQ =$	$m\angle KQP =$

2. The corresponding angles of $\triangle KJL$ and $\triangle KPQ$ are _____.

3. Complete the table.

Sides	Ratio
Determine the ratio of JK to PK .	
Determine the ratio of KL to KQ .	
Determine the ratio of JL to PQ .	

4. The corresponding sides of $\triangle KJL$ and $\triangle KPQ$ are _____.

5. $\triangle KJL$ is similar to $\triangle$ _____.
6. Describe a sequence of transformations that can be used to map $\triangle KJL$ onto $\triangle KPQ$.

Complete each relationship for $\triangle KJL$ and $\triangle KPQ$.

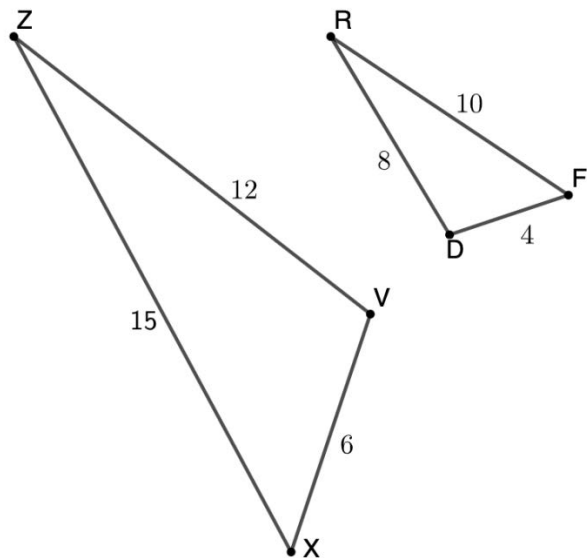
7. $\frac{PK}{JL} = \frac{\quad}{\quad}$

8. $\frac{JK}{QP} = \frac{\quad}{\quad}$

9. $\frac{KQ}{KJ} = \frac{\quad}{\quad}$

10. Observing that $\frac{JL}{PQ} \neq \frac{LQ}{QK}$, your classmate Moriah claims that $\triangle KJL$ and $\triangle KPQ$ are not similar. If her logic is faulty, explain why.

$\triangle ZVX$ and $\triangle RFD$ are shown.



1. Complete the table.

Measurements of $\triangle ZVX$	Measurements of $\triangle RFD$	Ratio
$ZX =$	$RF =$	$\frac{ZX}{RF} =$
$XV =$	$FD =$	$\frac{XV}{FD} =$
$VZ =$	$DR =$	$\frac{VZ}{DR} =$

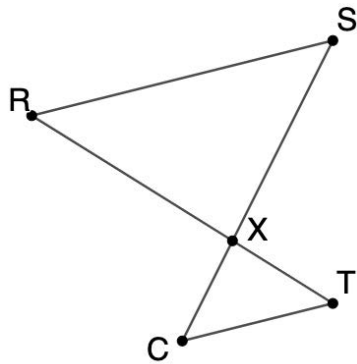
2. What similarity theorem can be applied to show that the triangles are similar?

3. Write the similarity statement.

4. Complete each.

$\frac{ZV}{FD} = \frac{VX}{DR}$
 $\frac{RF}{VX} = \frac{ZX}{DR}$
 $\frac{XZ}{DR} = \frac{VZ}{FD}$

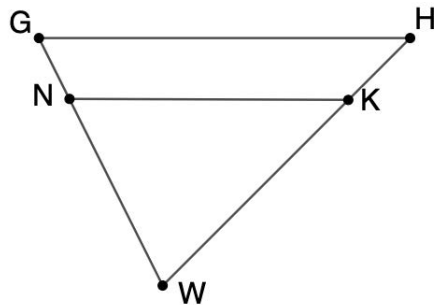
$\triangle RSX$ and $\triangle XTC$ are shown where $\angle RSX \cong \angle TCX$.



Complete the two-column proof.

	Statement	Reason
	$\angle RSX \cong \angle TCX$	Given
5.	$\angle SXR \cong \angle CXT$	
6.	$\triangle RSX \sim \triangle \underline{\hspace{2cm}}$	

$\triangle GWH$ and $\triangle NWK$ are shown.



Given: $WN = \frac{2}{3}WG$ and $WK = \frac{2}{3}WH$
Prove: $\triangle GWH \sim \triangle NWK$

	Statement	Reason
	$WN = \frac{2}{3}WG$ and $WK = \frac{2}{3}WH$	Given
7.	$\underline{\hspace{1cm}} = \frac{2}{3}$ and $\underline{\hspace{1cm}} = \frac{2}{3}$	Division Property of Equality
8.	$\frac{WN}{WG} = \frac{WK}{WH}$	
9.	$\angle GWH \cong \angle NWK$	
10.	$\triangle GWH \sim \triangle NWK$	

Geometry
Unit 6
Lesson 9 Homework

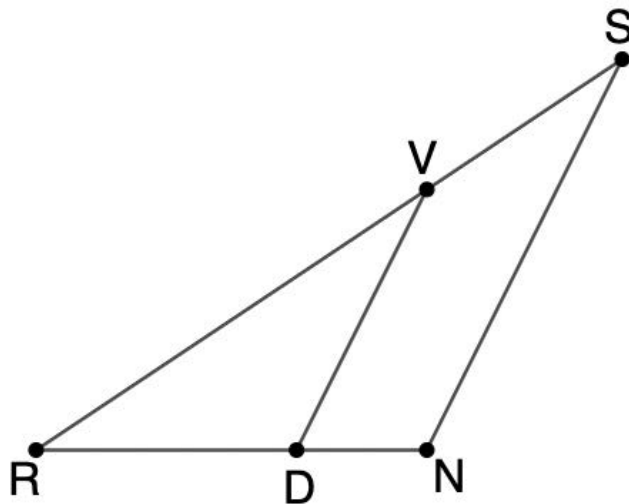
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$\triangle RSN$ with $\overline{VD}$ is shown.

Given: $RV = \frac{2}{3}RS$ and
 $RD = \frac{2}{3}RN$

Prove: $\triangle RSN \sim \triangle RVD$

Complete the two-column proof with the provided statements and reasons.



Statement	Reason
$RV = \frac{2}{3}RS$ and $RD = \frac{2}{3}RN$	Given
1.	
2.	
3.	
4.	

$\triangle RSN \sim \triangle RVD$

SAS ~

$\frac{RV}{RS} = \frac{RD}{RN}$

Transitive Property of Equality

$\angle VRD \cong \angle SRN$

Division Property of Equality

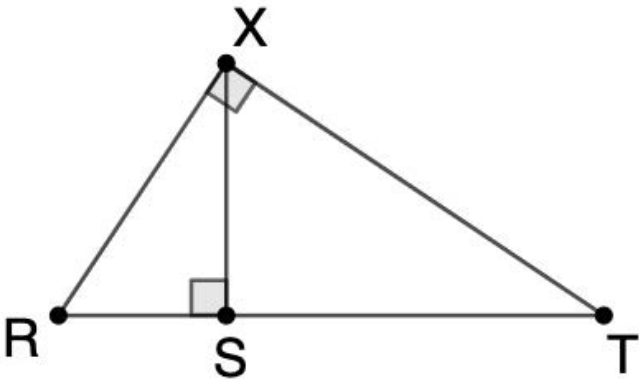
$\frac{RV}{RS} = \frac{2}{3}$ and $\frac{RD}{RN} = \frac{2}{3}$

Reflexive Property of Congruence

$\triangle RXT$ with $\overline{XS}$ is shown.

Given: $\overline{XS} \perp \overline{RT}$ and
 $\overline{RX} \perp \overline{XT}$

Prove: $\triangle XTR \sim \triangle SXR$



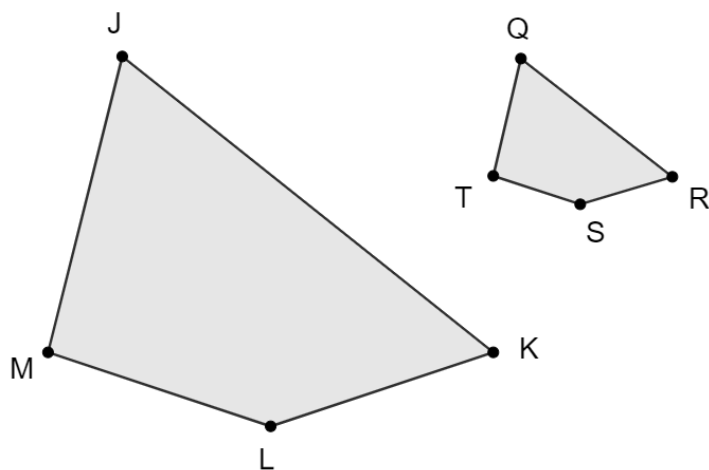
Complete the two-column proof with the provided statements and reasons.

	Statement	Reason
	$\overline{XS} \perp \overline{RT}$ and $\overline{RX} \perp \overline{XT}$	Given
5.		
6.		
7.		
8.		
9.		
10.		

$\angle XSR \cong \angle TXR$	AA ~
$\triangle XTR \sim \triangle SXR$	Definition of congruence
$\angle XRS \cong \angle TRX$	Definition of perpendicular lines
$m\angle XSR = m\angle TXR$	Reflexive Property of Congruence
$m\angle XSR = 90^\circ$ and $m\angle TXR = 90^\circ$	Transitive Property of Equality
$\angle XSR$ and $\angle TXR$ are right angles	Definition of right angles

Geometry
Unit 6
Lesson 10 Homework

Name _____
Date _____
Period _____

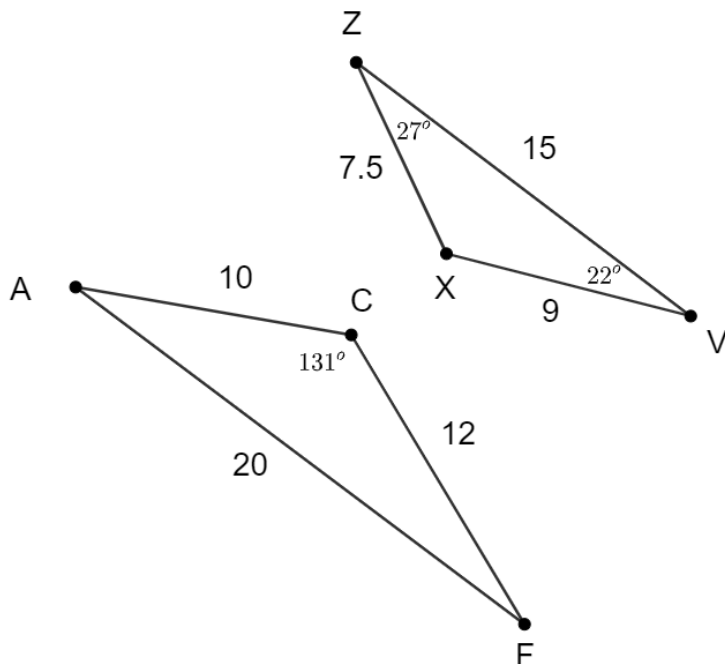


Fill in the table given that $JKL \sim QRS$.

	Corresponding Angles	Corresponding Sides
1.	$\angle JML \cong \angle$	$\frac{JM}{SR} = \frac{ML}{QR}$
2.	$\angle TQR \cong \angle$	$\frac{SR}{QR} = \frac{QR}{QR}$
3.	$\angle TSR \cong \angle$	$\frac{JK}{TS} = \frac{QR}{QR}$
4.	$\angle JKL \cong \angle$	$\frac{TS}{QR} = \frac{LK}{QR}$

5. If $ML = 6.5$, $TS = 2.5$, and $QR = 5$, determine JK .

$\triangle ACF$ and $\triangle VXZ$ are shown. $AC = 10$, $CF = 12$, $AF = 20$, $VX = 9$, $XZ = 7.5$, $ZV = 15$, $m\angle ACF = 131^\circ$, $m\angle XZV = 27^\circ$, and $m\angle ZVX = 22^\circ$.



6. What similarity theorem or postulate could be used to prove the triangles are similar?

7. Write a similarity statement for $\triangle ACF$ and $\triangle VXZ$.

Complete the table.

	Angle Measures	Side Proportions
8.	$m\angle CAF =$	$\frac{AC}{ZX} =$
9.	$m\angle CFA =$	$\frac{CF}{XV} =$
10.	$m\angle ZXV =$	$\frac{FA}{VZ} =$

Geometry

Unit 6

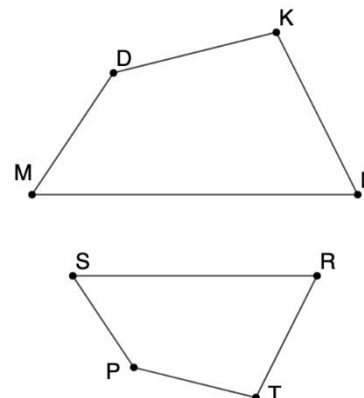
Lesson 11 Homework

Name _____

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$DKLM$ and $PTRS$ are shown, where $DKLM \sim PTRS$ with $MD = 20$, $DK = 6z + 6$, $KL = 8w + 1$, $LM = 40$, $PT = 12$, $TR = 4w - 2$, and $RS = 16$.



- Determine the value of each variable.

$$w = \underline{\hspace{2cm}}$$

$$z = \underline{\hspace{2cm}}$$

- Determine the value of each side.

$$DK = \underline{\hspace{2cm}}$$

$$SP = \underline{\hspace{2cm}}$$

$$TR = \underline{\hspace{2cm}}$$

- What is the perimeter of $DKLM$?

Consider additional measurements of angles for $DKLM$ and $PTRS$ where $m\angle DKL = 150^\circ$, $m\angle MDK = (15x - 25)^\circ$, $m\angle SPT = (12x + 5)^\circ$, $m\angle DML = (5y - 5)^\circ$, $m\angle PSR = (4y + 4)^\circ$.

- Determine the value of each variable.

$$x = \underline{\hspace{2cm}}$$

$$y = \underline{\hspace{2cm}}$$

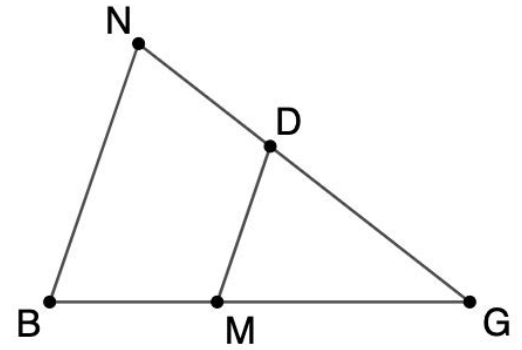
- Determine the value of each angle.

$$m\angle MDK = \underline{\hspace{2cm}}$$

$$m\angle PTR = \underline{\hspace{2cm}}$$

$$m\angle PSR = \underline{\hspace{2cm}}$$

$\triangle NBG$ and $\triangle DMG$ are shown, where $\triangle NBG \sim \triangle DMG$ with $DM = 8$, $NB = 32$, $DG = 7$, $MG = 2y - 8$, and $BG = 4y + 4$.



6. What is the scale factor from $\triangle DMG$ to $\triangle NBG$?

7. What is the value of the variable, y ?

8. What is the length of BG ?

Consider additional measurements of angles for $\triangle NBG$ and $\triangle DMG$ where $m\angle GNB = (8z + 3)^\circ$, $m\angle DMG = (7w - 9)^\circ$, $m\angle NBG = (6w + 3)^\circ$, $m\angle GDM = (10z - 15)^\circ$.

9. Determine the value of each variable.

$w = \underline{\hspace{2cm}}$

$z = \underline{\hspace{2cm}}$

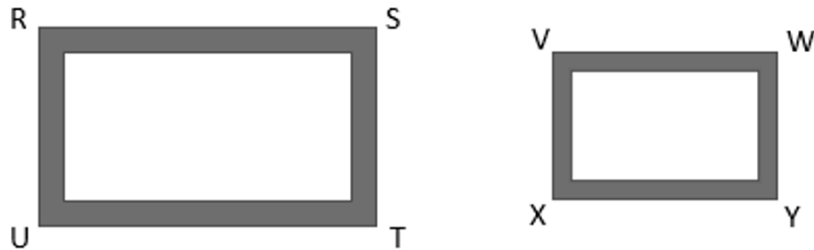
10. Determine the value of each angle.

$m\angle GNB = \underline{\hspace{2cm}}$

$m\angle GBN = \underline{\hspace{2cm}}$

$m\angle DGM = \underline{\hspace{2cm}}$

Hillary is custom building frames for a set of family photos. She has designed two similar rectangular frames, $RSTU$ and $VWYX$. The diagram shows the two different sized frames, where $WY = 12$ inches (in.), $VW = (3d + 11)$ in., $ST = 16$ in., and $RS = (8d - 4)$ in.



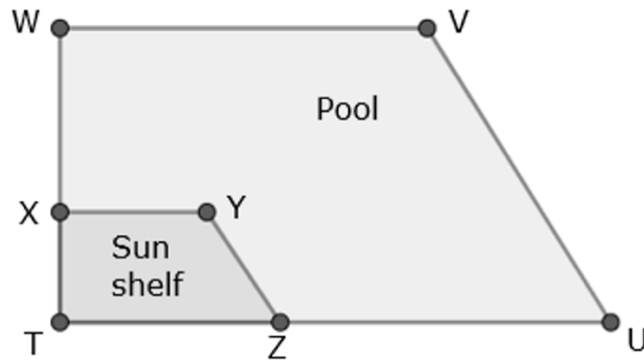
1. What is the value of d ?
2. What is the area of $RSTU$? Include units.
3. What is the perimeter of $VWYX$? Include units.

Hillary created a third similar rectangular frame, $LMNO$, shown below, where $LM = 12$ in.



4. What is the length of $\overline{LO}$? Include units.
5. What is the perimeter, in inches, of $LMNO$?

Laurel and her family are having a pool built in the backyard. The pool, $TUVW$, is shaped like a trapezoid, with a smaller similar trapezoid, $TZYX$, for a sun shelf. A diagram of the pool is shown, where $TU = (5a - 1)$ feet (ft), $TZ = (2a - 2)$ ft, $ZY = 5$ ft, $UV = 15$ ft, $VW = (10b + 1)$ ft, and $YX = (3b + 1)$ ft.



6. Determine the value of each variable:

$$a = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$

7. Determine the value of each side:

$$WV = \underline{\hspace{2cm}}$$

$$TZ = \underline{\hspace{2cm}}$$

8. If $WV = WT$, what is the perimeter, in feet, of the pool?

Some measurements of angles of $TUVW$ and $TZYX$ are given.

$$m\angle VUZ = (12c - 6)^\circ, m\angle YZT = (10c + 5)^\circ, m\angle XTZ = 90^\circ,$$

$$m\angle TXY = (4d + 10)^\circ, m\angle TWV = (5d - 10)^\circ$$

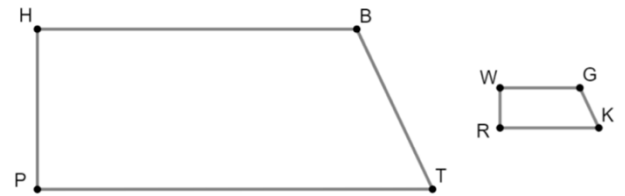
9. Determine the value of each variable:

$$c = \underline{\hspace{2cm}}$$

$$d = \underline{\hspace{2cm}}$$

10. What is the measure of $\angle XYZ$?

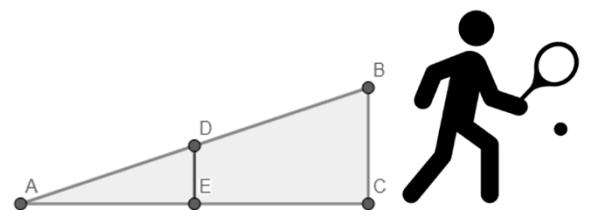
Sheena, a party planner, is designing a banner for a birthday party. Her client wants two similar banners: a large banner, $HBTP$, for the main entrance, and a similar sized banner, $WGKR$, for a photo booth.



Sheena knows that $PT = 55$ inches (in), $HP = 3x + 2$ in, $WG = 8$ in, $WR = 0.5x + 1$ in, $BT = 5$ in, and $RK = 11$ in.

1. What scale factor will map $HBTP$ to $WGKR$?
2. What is the perimeter of $HBTP$, the main entrance banner?
3. What is the perimeter of $WGKR$, the photo booth banner?

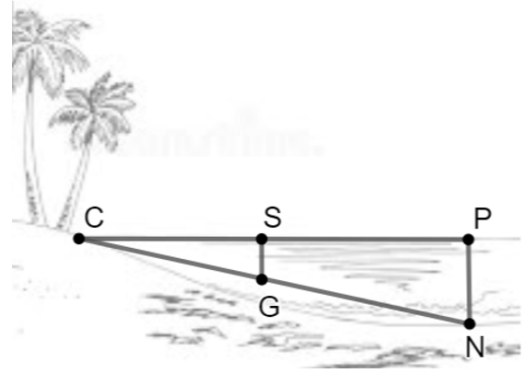
Venus is doing a tennis drill where she wants to hit the ball so that it will just pass over the net, $\overline{DE}$, and land 5 meters (m) away from the base of the net.



4. Given that $\triangle ADE \sim \triangle ABC$, $AE = 5$ m, $DE = 0.92$ m, and $EC = 10$ m, what is the height, represented by $\overline{BC}$, of the tennis racket?
5. What scale factor will map $\triangle ABC$ to $\triangle ADE$?

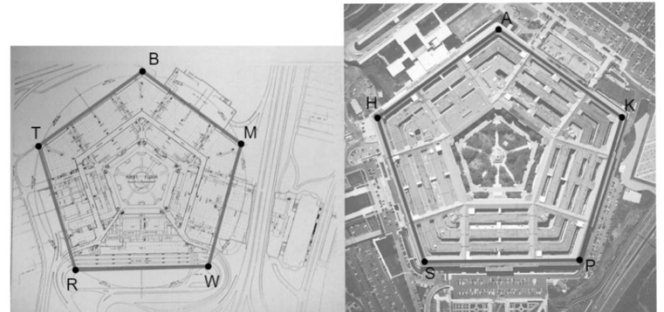
A beach front property owner is building a dock which extends the length of $\overline{CP}$ into the lake. $\overline{PN}$ and $\overline{SG}$ are the two main support posts for the dock.

$\triangle CSG \sim \triangle CPN$, $CS = 8$ feet (ft), $SG = (2x - 4)$ ft,
 $PN = (5x + 8)$ ft, and $CP = 32$ ft.



6. What scale factor will map $\triangle CPN$ to $\triangle CSG$?
7. What is the depth of the longer support post, represented by $\overline{PN}$?
8. What is the depth of the shorter support post, represented by $\overline{SG}$?

Cesar wants to create a scale model of the Pentagon for his social studies project. A copy of the blueprint, $BMWRT$, will guide him in building a similar scale model, $AKPSH$. The blueprint's sides, $BMWRT$, are $\frac{1}{5}$ of the scale model Pentagon, $AKPSH$. $TB = \left(\frac{2}{3}x + 4\right)$ centimeters (cm) and $HA = (6x + 4)$ cm.



9. What is the perimeter of the Pentagon in the blueprint, $BMWRT$?
10. What is the perimeter of the scale model Pentagon, $AKPSH$?